

AgentIQ Datathon-FINAL REPORT

MANDILENS WITH AI CHATBOT(Graph integrated)

Power BI Dashboard Link: [CLICK ME](#)

Chatbot LINK: [CLICK ME](#)

1. Introduction

Agriculture is an important part of India's economy, and mandis play an important role in the buying and selling of agricultural produce. Mandis generate useful information about crop arrivals, prices, transportation and weather. However, this information can be difficult to understand when it is spread across different datasets.

We developed MandiLens to bring this information together and make it easier to explore. The project provides users with two simple ways to understand mandi data. The Power BI dashboard presents market information through interactive charts, comparisons and filters, while the AI chatbot allows users to ask questions about the data in natural language.

Through MandiLens, users can explore information such as crop prices, mandi arrivals, market trends and relationships between market activity and weather conditions. The aim is to make agricultural data easier to access and understand, helping users gain useful insights without having to work directly with raw datasets.

2. Problem Statement

Agricultural mandi data contains valuable information about crop prices, arrivals, transportation and weather conditions, but this information is not always easy to access or interpret. Users may have to look through large datasets or different sources to find simple answers, such as which mandi has the highest arrivals or how crop prices have changed over time.

This creates a gap between the availability of agricultural data and its practical use. There is a need for a simple platform that can bring relevant information together and present it in an understandable form.

MandiLens addresses this problem by providing an interactive dashboard and an AI-based chatbot through which users can explore mandi data, identify trends and obtain useful information more easily.

3. Objectives

The main objectives of **MandiLens** are:

- To bring important agricultural mandi information together in one platform.
- To make mandi prices, crop arrivals and market trends easier to understand.
- To allow users to explore data through an interactive **Power BI dashboard**.
- To provide an **AI chatbot** through which users can ask questions about the data in natural language.
- To help users identify useful patterns and comparisons in mandi and agricultural data.
- To make data-based information more accessible for better understanding and decision-making.

8. Data Cleaning and Preparation

Before using the data in MandiLens, we cleaned and prepared the datasets to make the information more reliable and consistent. We checked the datasets for missing values, duplicate records and inconsistent entries and handled them wherever required.

We also standardized important fields such as dates, crop names, mandi names and identifiers so that information from different datasets could be used together correctly. Relevant datasets were then connected using common fields such as mandi, crop and date.

During the cleaning process, we made sure that missing information was not treated as zero and that unavailable mandi information was not filled by guessing. This helped us maintain the original meaning of the data and avoid misleading results.

The cleaned data was then used for the Power BI dashboard and the AI chatbot, allowing both parts of MandiLens to work with a consistent set of agricultural market information.

5. Benefits

MandiLens provides a simpler way to work with agricultural market information. It can help users:

- Understand mandi prices and crop arrivals more easily.
- Compare different crops and mandis.

- Identify market trends and patterns.
- Find information without manually searching through large datasets.
- View information through charts, tables and simple explanations.
- Use natural-language questions to explore agricultural data.

6. Key Insights from the Dashboard

The dashboard gives a quick view of the agricultural market and helps identify important patterns in the available data. Some of the main observations are:

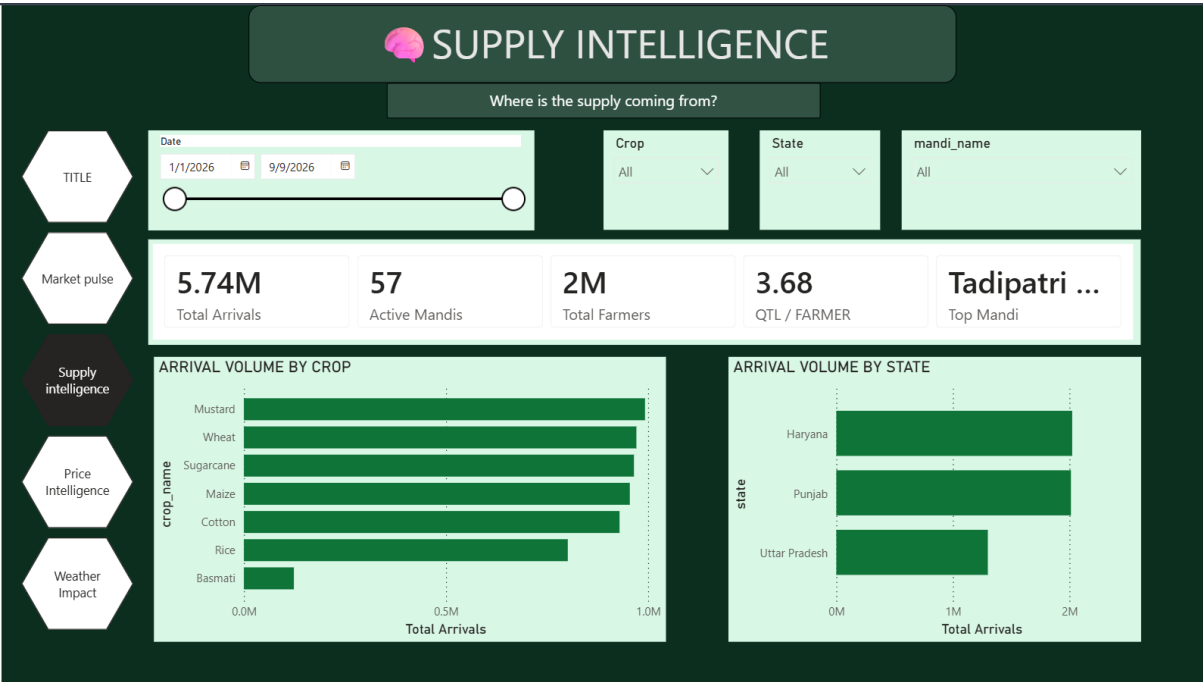
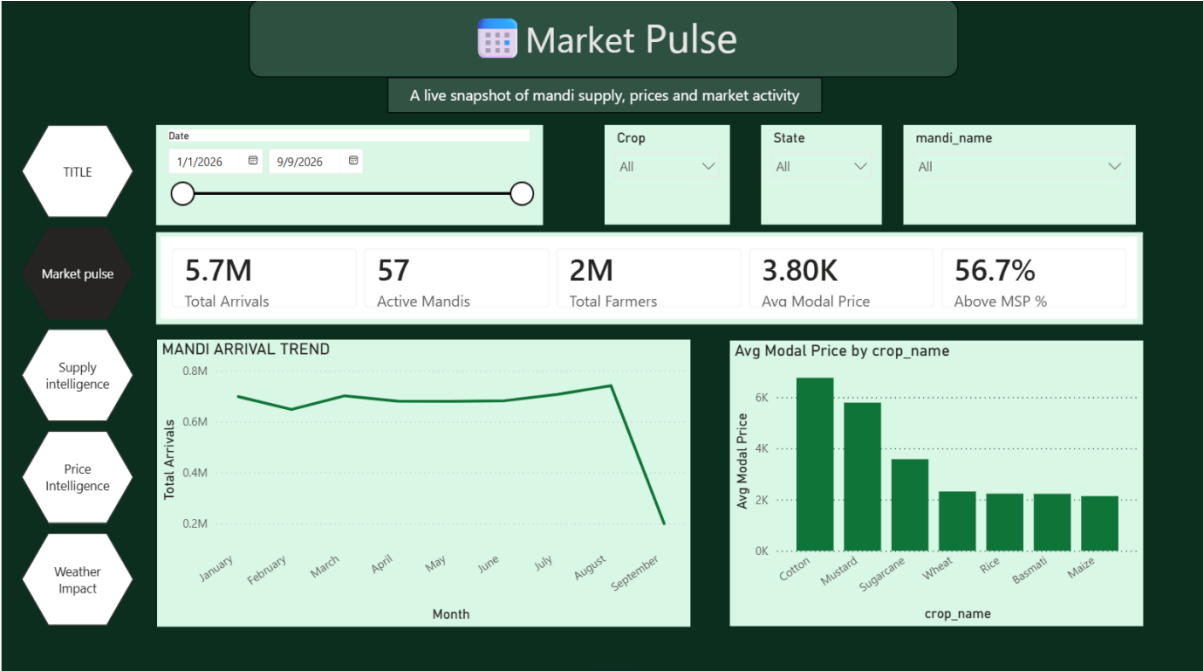
- The dataset covers 57 active mandis and around 5.7 million quintals of arrivals.
- Mustard and wheat account for some of the highest arrival volumes among the crops shown.
- Punjab and Haryana show the highest arrival volumes compared with Uttar Pradesh in the dashboard.
- The average market price is around ₹3,800, while the average MSP is around ₹3,730.
- Around 56.7% of the recorded prices are above MSP, giving an overall view of price realization.
- Mandi arrivals remain fairly consistent through most of the months shown, with a noticeable fall in September because the available data ends on September 9.
- The weather section allows users to view rainfall, temperature and humidity alongside mandi activity, making it easier to look for possible relationships between weather and market supply.

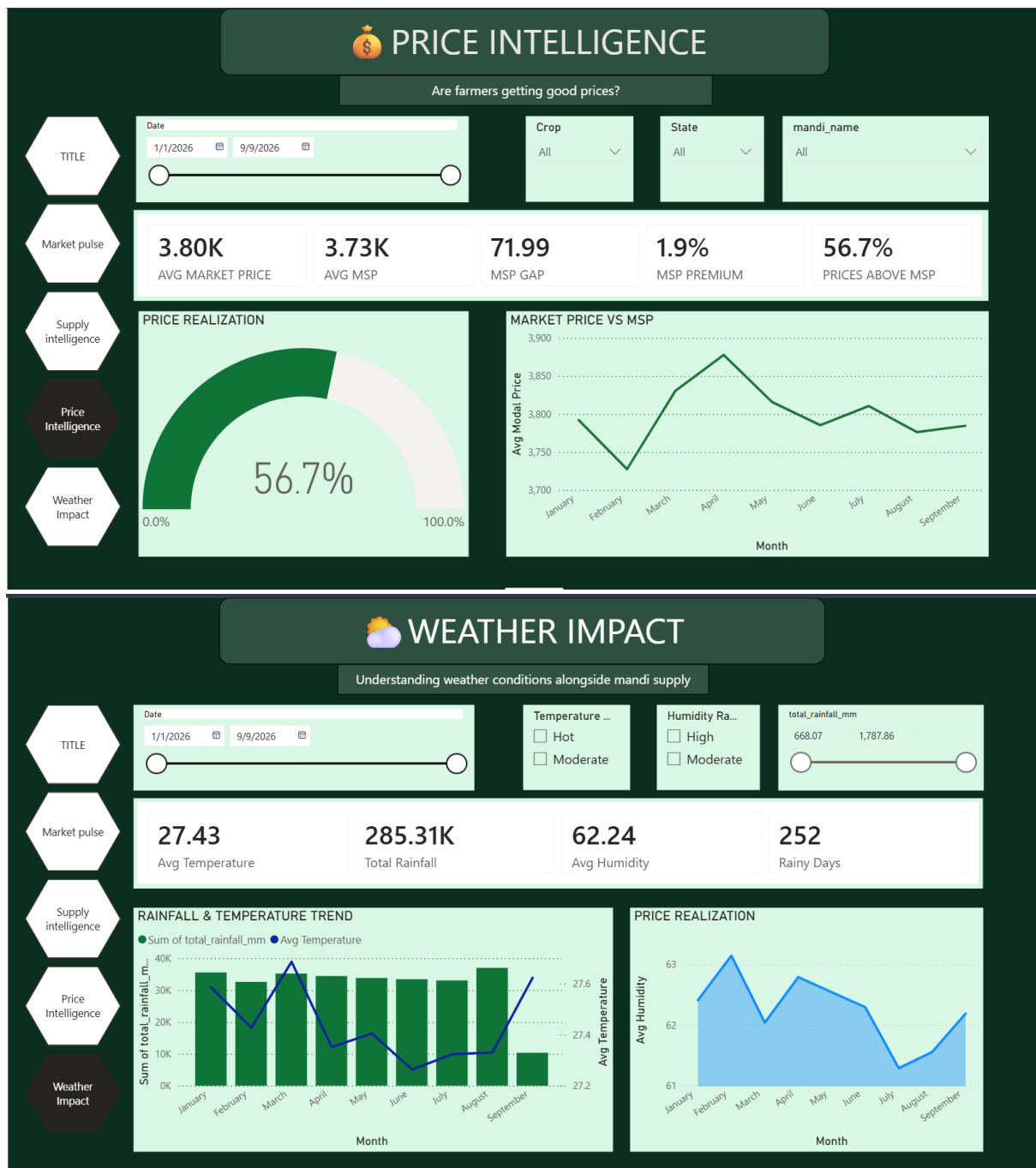
7. Power BI Dashboard

The Power BI dashboard is designed to provide a clear overview of agricultural mandi activity. It allows users to explore information about crop arrivals, prices and other market-related factors using interactive visuals and filters.

Users can compare different mandis and crops, observe changes over time and identify patterns in the available data. The dashboard presents the information visually so that important trends can be understood more quickly than by looking at raw data.

Power BI Dashboard Link: [CLICK ME](#)





8. AI Chatbot

The AI chatbot provides a simple way for users to interact with the agricultural data. Instead of manually searching through datasets or dashboards, users can ask questions in natural language, such as questions about crop prices, mandi arrivals or market trends.

The chatbot processes the user's question and provides an answer based on the available mandi data. Depending on the question, it can also present the result through suitable visualizations such as charts or tables. This makes the information easier to understand and allows users to explore the data through a conversational interface.

AI Chatbot Link: [CLICK ME](#)



9. Conclusion

MandiLens brings agricultural market information into one place and makes it easier to understand. Through the Power BI dashboard and AI chatbot, users can explore mandi activity, prices, crop arrivals and weather-related information without dealing directly with large datasets.

The project shows how combining data visualization with an AI-based interface can make agricultural information more accessible. MandiLens can serve as a useful starting point for understanding market conditions and supporting data-based decisions.

11. Future Scope

MandiLens can be further improved by adding real-time mandi prices and arrivals, more locations and crops, and forecasting features for prices and supply. It could also provide personalized alerts for major price changes, weather conditions or mandi activity.

In the future, the platform could be made available as a mobile or web application so that farmers and other users can access the information more easily from anywhere.